

Write a Program to Perform Log Analysis using Map Reduce

Step 1 — Start Hadoop

Run:

```
start-dfs.sh
```

Then:

```
start-yarn.sh
```

Check:

```
jps
```

You should normally see:

NameNode

DataNode

SecondaryNameNode

ResourceManager

NodeManager

The exact list can vary slightly depending on your configuration.

```
hadoop@Suhel-PC:~$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [Suhel-PC]
hadoop@Suhel-PC:~$ start-yarn.sh
Starting resourcemanager
Starting nodemanagers
hadoop@Suhel-PC:~$ jps
1636 NodeManager
949 DataNode
1495 ResourceManager
807 NameNode
2044 Jps
1212 SecondaryNameNode
```

Step 2 — Create a working directory

Create a directory for this experiment:

```
mkdir -p ~/web_log_lab
```

```
cd ~/web_log_lab
```

Check:

```
pwd
```

You should get:

```
/home/hadoop/web_log_lab
```

```
1212 secondarynamenode
hadoop@Suhe1-PC:~$ mkdir -p ~/web_log_lab
cd ~/web_log_lab
hadoop@Suhe1-PC:~/web_log_lab$ pwd
/home/hadoop/web_log_lab
```

Step 3 — Create log_data.txt

Run:

```
nano log_data.txt
```

Paste this data.

```
96.7.8.17 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
96.7.1.14 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
96.7.2.14 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
192.168.1.1 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
96.7.4.16 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
91.75.5.14 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
98.21.6.14 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
162.15.16.1 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
8.8.8.8 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
10.99.99.247 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
96.7.1.14 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
96.7.1.14 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
```

Step 4 — Check the number of records

Run:

```
wc -l log_data.txt
```

You should get:

12 log_data.txt

```
/home/hadoop/web_log_lab
hadoop@Suhe1-PC:~/web_log_lab$ nano log_data.txt
hadoop@Suhe1-PC:~/web_log_lab$ cat log_data.txt
96.7.8.17 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
96.7.1.14 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
96.7.2.14 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
192.168.1.1 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
96.7.4.16 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
91.75.5.14 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
98.21.6.14 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
162.15.16.1 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
8.8.8.8 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
10.99.99.247 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
96.7.1.14 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
96.7.1.14 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
hadoop@Suhe1-PC:~/web_log_lab$ wc -l log_data.txt
12 log_data.txt
```

Step 5 — Create the Java program

Create:

```
nano LogAnalysis.java
```

Paste the below code:

```
import java.io.IOException;

import org.apache.hadoop.conf.Configuration;

import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;

import org.apache.hadoop.io.LongWritable;

import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;

import org.apache.hadoop.mapreduce.Mapper;

import org.apache.hadoop.mapreduce.Reducer;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;

import org.apache.hadoop.mapreduce.lib.input.TextInputFormat;
```

```
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
import org.apache.hadoop.mapreduce.lib.output.TextOutputFormat;
```

```
public class LogAnalysis {
```

```
    public static class LogMapper
```

```
        extends Mapper<LongWritable, Text, Text, IntWritable> {
```

```
        private static final IntWritable one = new IntWritable(1);
```

```
        private final Text ipAddress = new Text();
```

```
        @Override
```

```
        public void map(LongWritable key,
```

```
                        Text value,
```

```
                        Context context)
```

```
        throws IOException, InterruptedException {
```

```
        String line = value.toString().trim();
```

```
        // Ignore empty lines
```

```
        if (line.isEmpty()) {
```

```
            return;
```

```
        }
```

```
        // Ignore header if present
```

```
        if (line.startsWith("IP Address")) {
```

```
            return;
```

```

    }

    // First field is the IP address
    String[] parts = line.split("\\s+");

    if (parts.length >= 1) {
        ipAddress.set(parts[0]);
        context.write(ipAddress, one);
    }
}

}

}

public static class LogReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {

    private final IntWritable result = new IntWritable();

    @Override
    public void reduce(Text key,
        Iterable<IntWritable> values,
        Context context)
        throws IOException, InterruptedException {

        int sum = 0;

        for (IntWritable val : values) {
            sum += val.get();
        }
    }
}

```

```
        result.set(sum);
        context.write(key, result);
    }
}
```

```
public static void main(String[] args) throws Exception {
```

```
    if (args.length != 2) {
        System.err.println(
            "Usage: LogAnalysis <input path> <output path>"
        );
        System.exit(1);
    }
```

```
    Configuration conf = new Configuration();
```

```
    Job job = Job.getInstance(conf, "Web Log Analysis");
```

```
    job.setJarByClass(LogAnalysis.class);
```

```
    job.setMapperClass(LogMapper.class);
```

```
    // Combiner reduces intermediate data
```

```
    job.setCombinerClass(LogReducer.class);
```

```
    job.setReducerClass(LogReducer.class);
```

```

job.setOutputKeyClass(Text.class);
job.setOutputValueClass(IntWritable.class);

job.setInputFormatClass(TextInputFormat.class);
job.setOutputFormatClass(TextOutputFormat.class);

FileInputFormat.addInputPath(
    job,
    new Path(args[0])
);

FileOutputFormat.setOutputPath(
    job,
    new Path(args[1])
);

System.exit(
    job.waitForCompletion(true) ? 0 : 1
);
}
}

```

Step 6 — Compile the Java program

This is where Hadoop 3.3.6 makes things convenient.

Run:

```
mkdir -p classes
```

Then compile using Hadoop's classpath:

```
javac -classpath "$(hadoop classpath)" -d classes LogAnalysis.java
```

```
hadoop@Suhel-PC:~/web_log_lab$ nano LogAnalysis.java
hadoop@Suhel-PC:~/web_log_lab$ mkdir -p classes
hadoop@Suhel-PC:~/web_log_lab$ javac -classpath "$(hadoop classpath)" -d classes LogAnalysis.java
hadoop@Suhel-PC:~/web_log_lab$ find classes -type f
classes/LogAnalysis$LogReducer.class
classes/LogAnalysis$LogMapper.class
classes/LogAnalysis.class
```

Check:

find classes -type f

Step 7 — Create the JAR file

Run:

```
jar -cvf LogAnalysis.jar -C classes .
```

Check:

```
ls -lh LogAnalysis.jar
```

Step 8 — Check the JAR contents

Run:

```
jar tf LogAnalysis.jar
```

```
hadoop@Suhel-PC:~/web_log_lab$ jar -cvf LogAnalysis.jar -C classes .
added manifest
adding: LogAnalysis$LogReducer.class(in = 1739) (out= 738)(deflated 57%)
adding: LogAnalysis$LogMapper.class(in = 2018) (out= 858)(deflated 57%)
adding: LogAnalysis.class(in = 1867) (out= 970)(deflated 48%)
hadoop@Suhel-PC:~/web_log_lab$ ls -lh LogAnalysis.jar
-rw-rw-r-- 1 hadoop hadoop 3.3K Oct  1 03:32 LogAnalysis.jar
hadoop@Suhel-PC:~/web_log_lab$ jar tf LogAnalysis.jar
META-INF/
META-INF/MANIFEST.MF
LogAnalysis$LogReducer.class
LogAnalysis$LogMapper.class
LogAnalysis.class
hadoop@Suhel-PC:~/web_log_lab$
```

Step 9 — Create HDFS input directory

Create:

```
hdfs dfs -mkdir -p /weblog
```

Check:

```
hdfs dfs -ls /
```

You should see:

```
/weblog
```

Step 10 — Upload the input file to HDFS

Your local file is:

```
/home/hadoop/web_log_lab/log_data.txt
```

Upload:

```
hdfs dfs -put -f log_data.txt /weblog/
```

Check:

```
hdfs dfs -ls /weblog
```

```

hadoop@Suhel-PC:~/web_log_lab$ hdfs dfs -mkdir -p /weblog
hadoop@Suhel-PC:~/web_log_lab$ hdfs dfs -ls /
Found 4 items
drwxr-xr-x - hadoop supergroup          0 2026-09-17 04:13 /WordCCount
drwxr-xr-x - hadoop supergroup          0 2026-09-24 05:09 /input
drwxr-xr-x - hadoop supergroup          0 2026-09-24 05:11 /output
drwxr-xr-x - hadoop supergroup          0 2026-10-01 03:33 /weblog
hadoop@Suhel-PC:~/web_log_lab$ /home/hadoop/web_log_lab/log_data.txt
-bash: /home/hadoop/web_log_lab/log_data.txt: Permission denied
hadoop@Suhel-PC:~/web_log_lab$ sudo /home/hadoop/web_log_lab/log_data.txt
sudo: I'm sorry hadoop. I'm afraid I can't do that
hadoop@Suhel-PC:~/web_log_lab$ hdfs dfs -put -f log_data.txt /weblog/
hadoop@Suhel-PC:~/web_log_lab$ hdfs dfs -ls /weblog
Found 1 items
-rw-r--r--  1 hadoop supergroup          883 2026-10-01 03:33 /weblog/log_data.txt
hadoop@Suhel-PC:~/web_log_lab$ hdfs dfs -cat /weblog/log_data.txt
96.7.8.17 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
96.7.1.14 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
96.7.2.14 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
192.168.1.1 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
96.7.4.16 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
91.75.5.14 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
98.21.6.14 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
162.15.16.1 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
8.8.8.8 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
10.99.99.247 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
96.7.1.14 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433
96.7.1.14 [24/Apr/2011:04:20:11 -0400] "GET /cat.jpg HTTP/1.1" 200 12433

```

Step 11 — Verify the HDFS file

Run:

```
hdfs dfs -cat /weblog/log_data.txt
```

Step 12 — Run the MapReduce program

```
hadoop jar LogAnalysis.jar LogAnalysis /weblog/log_data.txt /loganalysisresult
```

```

hadoop@Suhel-PC:~/web_log_lab$ hadoop jar LogAnalysis.jar LogAnalysis /weblo
g/log_data.txt /loganalysisresult

```

```
Combine input records=12
Combine output records=10
Reduce input groups=10
Reduce shuffle bytes=173
Reduce input records=10
Reduce output records=10
Spilled Records=20
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=19
Total committed heap usage (bytes)=1177550848
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=883
File Output Format Counters
  Bytes Written=127
```

Step 13 — Check the output directory

Run:

```
hdfs dfs -ls /loganalysisresult
```

Typically you will see:

```
_SUCCESS
```

```
part-r-00000
```

Step 14 — Display the result

Run:

```
hdfs dfs -cat /loganalysisresult/part-r-00000
```

```
hadoop@Suhel-PC:~/web_log_lab$ hdfs dfs -ls /loganalysisresult
Found 2 items
-rw-r--r-- 1 hadoop supergroup 0 2026-10-01 03:34 /loganalysisresult/_SUCCESS
-rw-r--r-- 1 hadoop supergroup 127 2026-10-01 03:34 /loganalysisresult/part-r-00000
hadoop@Suhel-PC:~/web_log_lab$ hdfs dfs -cat /loganalysisresult/part-r-00000
10.99.99.247 1
162.15.16.1 1
192.168.1.1 1
8.8.8.8 1
91.75.5.14 1
96.7.1.14 3
96.7.2.14 1
96.7.4.16 1
96.7.8.17 1
98.21.6.14 1
```